SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I s.prajith kumar_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit

assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent? ii. Characterize the operating environment. iii. What are the actions the agent can take? iv. How can one evaluate the performance of the agent?

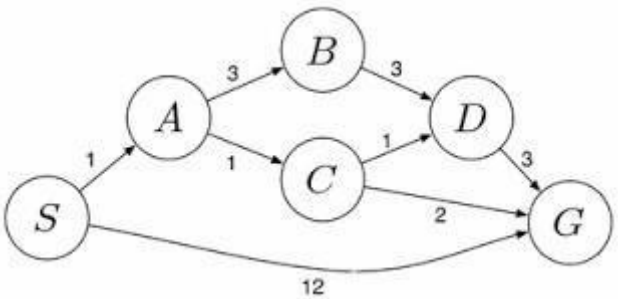
v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



1. Water Jug Problem (4-gallon and 3-gallon jugs)

Goal: Obtain exactly 2 gallons in the 4-gallon jug.

Step	4-Gallon Jug	3-Gallon Jug
Start	0	0
Fill 3-gallon jug	0	3
Pour 3 → 4	3	0
Fill 3-gallon jug again	3	3
Pour 3 → 4 until full	4	2
Empty 4-gallon jug	0	2
Pour remaining 2 gallons into 4-gallon jug	2	0

Result: The 4-gallon jug contains exactly 2 gallons.

2. Mars Rover as an Intelligent Agent

i. Percepts

- Camera images
- Rock/soil composition data
- Temperature readings
- Atmospheric measurements
- Position and terrain information **ii. Operating Environment**
- Partially observable
- Dynamic
- Sequential
- Stochastic

- Real-world environment

iii. Actions

- Move forward/backward
- Turn and navigate
- Collect samples
- Analyze rocks and soil
- Capture images
- Send data to Earth **iv. Performance Measure**
- Number of useful samples collected
- Accuracy of analysis
- Distance explored
- Energy efficiency
- Mission success rate

v. Suitable Agent Architecture

- **Model-Based Goal-Oriented Agent**

Reason: The rover must maintain an internal model of Mars, plan routes, avoid obstacles, and achieve exploration goals autonomously.

3. 8-Queens Problem Formulation

Problem Statement: Place 8 queens on a chessboard such that no two queens attack each other.

Problem Components

Initial State

- Empty 8 × 8 chessboard.

State Space

- All possible arrangements of queens on the board.

Actions

- Place a queen in a safe position in the next column.

Goal Test

- Eight queens are placed.
- No two queens share: ○ Same row ○ Same column ○ Same diagonal

Path Cost

- Each placement costs 1.

Search Strategy

- Depth First Search (DFS) with Backtracking.

Example Solution

Column : 1 2 3 4 5 6 7 8

Row : 1 5 8 6 3 7 2 4

This arrangement satisfies all constraints

4. OLA Cab Booking Application

Type of Problem-Solving Agent

- **Goal-Based Agent**

The user's goal is to reach a destination. The agent evaluates available cab types and selects the most suitable option.

Pseudocode

1. START
2. Input Source
3. Input Destination
4. Get available cabs
5. FOR each cab type
6. Check fare
7. Check availability
8. Check travel time
9. END FOR

10. Select best cab according to user preference
11. Book cab
12. Display booking confirmation
13. END

5 .Uniform Cost Search (UCS)

Problem Understanding

Determine the least-cost path between warehouse S and destination G using UCS.

Method Selection

Uniform Cost Search expands the node with the lowest cumulative cost first.

UCS Algorithm

1. UCS(Start, Goal)
2. Create Priority Queue
3. Insert Start Node with cost 0
4. WHILE Queue not empty
5. Remove node with minimum cost
6. IF Goal reached
7. Return path and cost
8. Expand node
9. Add neighbors with cumulative cost
10. END WHILE

Solution Process

1. Start at S with cost 0.
2. Expand the lowest-cost node.
3. Update cumulative path costs.
4. Continue until G is selected.
5. The first occurrence of G gives the optimal path.

Interpretation

Uniform Cost Search guarantees the optimal solution when all edge costs are non-negative.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem